Lecture 2: Gaussian Measures and Q-Wiener Processes on Hilbert Spaces

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This lecture develops Gaussian measures on Hilbert spaces, establishes the trace and spectral properties of the covariance operator, and introduces the canonical construction of the Q-Wiener process.

1. Foundation: Gaussian Measures on Hilbert Spaces

We establish our setting on a real, separable Hilbert space, denoted as H, equipped with an inner product $\langle \cdot, \cdot \rangle$.

We define a probability measure μ on the Borel σ-algebra $(H, \mathcal{B}(H))$ to be a Gaussian measure if and only if its characteristic function satisfies the following equality for all $h \in H$:

$$\int_H e^{i\langle h,u\rangle}\, \mu(du) = e^{im(h)-\frac{1}{2}\sigma^2(h)}.$$

2. Theorem 1.6: Characterization of the Measure

Theorem 1.6 gives us the explicit conditions for this measure.

Theorem 1 (Theorem 1.6)**.** The measure μ is Gaussian on H if and only if the characteristic function can be represented as

$$\int_H e^{i\langle h,u\rangle}\, \mu(du) = e^{i\langle m,h\rangle-\frac{1}{2}\langle Qh,h\rangle}.$$

From this theorem, we extract two fundamental components mapping from H:

The Mean: The mapping $h \mapsto m(h)$ is defined by the inner product $\langle m, h\rangle$, where $m \in H$.

The Variance/Covariance: The mapping $h \mapsto \sigma^2(h)$ is defined by the quadratic form $\langle Qh, h\rangle$.

For this to hold, the covariance operator Q must satisfy strict conditions: it must be a bounded linear operator, that is, $Q \in L(H)$, it must be symmetric, and it must be positive semidefinite. Most importantly, Q must be a trace-class operator, meaning it has a finite trace. We compute this trace with respect to a Complete Orthonormal System (CONS) (e_k) as follows:

$$\mathrm{tr}(Q) = \sum_{k=1}^{\infty} \langle Qe_k, e_k\rangle < \infty.$$

If the trace is finite for one CONS, it is finite for all such systems.

3. Application: The Brownian Bridge and the Inverse Dirichlet Laplacian

Before turning to the proof of Theorem 1.6, note a practical illustration involving a Brownian bridge process defined on the time interval $[0, T]$.

We consider the inverse Dirichlet Laplacian operator, denoted $(-\Delta_D)^{-1}$, acting on the space $L^2([0, T])$. The action of this operator on a function $g(t)$ is given by the integral equation

$$(-\Delta_D)^{-1} g(t) := -\int_0^T g(s) \left(s \wedge t - \frac{st}{T} \right) ds.$$

The kernel of this integral, $\left(s \wedge t - \frac{st}{T}\right)$, is the fundamental solution. By evaluating this fundamental solution at the boundaries $t = 0$ and $t = T$, one sees that it trivially equals 0, satisfying the required Dirichlet boundary conditions.

4. Sketch of the Proof for Theorem 1.6

Return now to the proof of Theorem 1.6, and outline the forward implication $\Rightarrow$.

First, we define a functional

$$\ell_h(u) := \langle h, u \rangle,$$

which is both linear and continuous. One must show that the mapping $h \mapsto m(h)$ is equal to the expectation of this functional, $E(\ell_h)$.

To rigorously establish this, one must prove that the operator norm is bounded, meaning

$$\sup_{\|h\| \leq 1} m(h) < \infty.$$

To this end, define closed subsets H_n within our Hilbert space by

$$H_n := \left\{ h \in H : \int_H |\ell_h(u)| \, \mu(du) \leq n \right\} \subset H.$$

The proof concludes by applying the Baire Category Theorem, utilizing the fact that the entire Hilbert space can be represented as the countable union of these closed subsets:

$$H = \bigcup_{n \geq 1} H_n.$$

As we already established the closed subsets H_n to investigate the boundedness of our mean functional, we now do the rigorous extraction of those bounds using the Baire Category Theorem, and then construct the covariance operator.

5. Fatou's Lemma and the Closedness of H_n

One must first confirm that the subsets H_n are closed. Let $h_n \to h$ be a convergent sequence in our Hilbert space H. By the continuity of the inner product, our linear functional converges pointwise:

$$|\ell_{h_n}(u)| \to |\ell_h(u)|.$$

Apply Fatou's Lemma to the integrals with respect to our measure μ:

$$\int_H |\ell_h(u)| \, \mu(du) \leq \liminf_{n \to \infty} \int_H |\ell_{h_n}(u)| \, \mu(du).$$

This inequality rigorously proves that the sets H_n are closed.

6. Applying the Baire Category Theorem

Since the entire Hilbert space is the countable union of these closed subsets,

$$H = \bigcup_{n \geq 1} H_n,$$

the Baire Category Theorem dictates that at least one of these subsets must have a non-empty interior. Denote this specific subset by H_{n_0}. Because its interior is non-empty, there exists a strictly positive radius $\varepsilon > 0$ and a center point $h_0 \in H$ such that the open ball

$$B_\varepsilon(h_0)$$

is completely contained within H_{n_0}.

7. Bounding the Mean Functional $m(h)$

To prove the functional is bounded, evaluate $h \in H$ subject to $\|h\| \leq 1$. Use an algebraic manipulation to scale and shift h directly into the defined ball $B_\varepsilon(h_0)$:

$$h = \frac{2}{\varepsilon}\left(\frac{\varepsilon}{2}h + h_0 - h_0\right).$$

Observe the term

$$\left(\frac{\varepsilon}{2}h + h_0\right).$$

Its distance from the center h_0 is exactly

$$\left\|\frac{\varepsilon}{2}h + h_0 - h_0\right\| = \frac{\varepsilon}{2}\|h\| \leq \frac{\varepsilon}{2} < \varepsilon.$$

Therefore,

$$\left(\frac{\varepsilon}{2}h + h_0\right) \in B_\varepsilon(h_0) \subset H_{n_0}.$$

By extracting this using the linearity of m, one has

$$|m(h)| = \frac{2}{\varepsilon}\left|m\left(\frac{\varepsilon}{2}h + h_0\right) - m(h_0)\right|.$$

Applying the triangle inequality separates the terms:

$$|m(h)| \leq \frac{2}{\varepsilon}\left|m\left(\frac{\varepsilon}{2}h + h_0\right)\right| + \frac{2}{\varepsilon}|m(h_0)|.$$

Because the vector $\left(\frac{\varepsilon}{2}h + h_0\right)$ resides within H_{n_0}, evaluate its integral representation:

$$\left|m\left(\frac{\varepsilon}{2}h + h_0\right)\right| = \left|\int_H \left\langle \frac{\varepsilon}{2}h + h_0, u\right\rangle \mu(du)\right| \leq \int_H \left|\left\langle \frac{\varepsilon}{2}h + h_0, u\right\rangle\right| \mu(du) \leq n_0.$$

This gives a hard upper bound. The mean functional is bounded on the unit ball, and therefore continuous.

8. Construction of the Covariance Operator Q

Concerning the covariance operator Q, one must define its action across the Hilbert space. For all $q, g \in H$, there exists an operator mapping $q \mapsto Q(q)$ such that the bilinear form $\langle Q(q), g\rangle$ is defined. Define this mapping $g \mapsto \langle Q(q), g\rangle$ explicitly as the integral of the centered inner products:

$$\int_H (\langle q, u\rangle - \langle q, m\rangle)(\langle g, u\rangle - \langle g, m\rangle)\, \mu(du).$$

Thus,

$$\langle Q(q), g\rangle = \int_H (\langle q, u\rangle - \langle q, m\rangle)(\langle g, u\rangle - \langle g, m\rangle)\, \mu(du).$$

This successfully establishes the existence and representation of the covariance operator Q, completing the structural requirements for the Gaussian measure.

As we already established the existence and basic positive semi-definite property of the covariance operator $\langle Qh, h\rangle \geq 0$, we now address the rigorous justification of the finite trace condition.

9. The Trace Bound Lemma

Lemma 1 (Trace bound lemma)**.** Assume a centered Gaussian measure,

$$\mu = N(0, Q).$$

Then the trace is bounded by the inverse square of the exponential expectation:

$$1 + \operatorname{tr}(Q) \leq \left(\int_H e^{-\frac{1}{2}\|x\|^2}\, \mu(dx) \right)^{-2} < \infty.$$

10. Side Remark: Girsanov's Theorem

Before continuing the proof of the trace bound lemma, record a brief side remark concerning absolute continuity of measures.

Theorem 2 (Girsanov's theorem)**.** Consider a path h on $[0, T]$ such that

$$h_t = \int_0^t \dot{h}_s\, ds$$

with finite kinetic energy, meaning

$$\int_0^T |\dot{h}_s|^2\, ds < \infty.$$

Then the shifted Wiener process $(B_t + h_t)_t$ induces a measure that is absolutely continuous with respect to the standard Wiener measure. This is an immediate consequence of Girsanov's Theorem.

11. Proof of the Trace Bound Lemma (Finite-Dimensional Case)

Proof. Return now to the trace bound proof. Restrict first to the finite-dimensional setting and set

$$\dim H = n < \infty \qquad \text{for } n \in \mathbb{N}.$$

Let $\sigma_1^2, \ldots, \sigma_n^2$ be the eigenvalues of Q. Define

$$\mu_n = N(0, Q_n)$$

as the distribution of μ with respect to the projection

$$\pi_n : H \to \mathbb{R}^n, \qquad h \mapsto (\langle h, e_1 \rangle, \ldots, \langle h, e_n \rangle).$$

This projection is defined over a Complete Orthonormal System (CONS) $(e_1, e_2, \ldots)$ consisting strictly of the eigenvectors of Q_n. Without loss of generality, assume strictly positive eigenvalues $\sigma_i^2 > 0$.

Because the measure $N(0, Q_n)$ is absolutely continuous with respect to the Lebesgue measure dx, write its density explicitly:

$$\frac{d\mu_n}{dx} = \frac{1}{\sqrt{(2\pi)^n \det Q_n}} \exp\left(-\frac{1}{2}\langle Q_n^{-1}x, x\rangle\right).$$

Utilizing the basis of eigenvectors, the quadratic form diagonalizes, allowing one to rewrite the density as

$$\frac{d\mu_n}{dx} = \frac{1}{\sqrt{(2\pi)^n \det Q_n}} \exp\left(-\sum_{i=1}^n \frac{x_i^2}{2\sigma_i^2}\right).$$

Now evaluate the integral from the lemma over $\mathbb{R}^n$:

$$\int_{\mathbb{R}^n} e^{-\frac{1}{2}\|x\|^2} \mu_n(dx) = \int_{\mathbb{R}^n} \frac{1}{\sqrt{(2\pi)^n \det Q_n}} \exp\left(-\sum_{i=1}^n \frac{(\sigma_i^2+1)x_i^2}{2\sigma_i^2}\right) dx.$$

This operates as a standard Gaussian integral. Evaluating it separates the terms into a product:

$$\int_{\mathbb{R}^n} e^{-\frac{1}{2}\|x\|^2} \mu_n(dx) = \frac{1}{\sqrt{(2\pi)^n \det Q_n}} \left((2\pi)^{n/2} \prod_{i=1}^n \frac{\sigma_i}{\sqrt{\sigma_i^2+1}}\right).$$

Knowing that

$$\det Q_n = \prod_{i=1}^n \sigma_i^2,$$

the normalization constants and the σ_i terms cancel precisely. One is left with the final closed-form result:

$$\int_{\mathbb{R}^n} e^{-\frac{1}{2}\|x\|^2} \mu_n(dx) = \prod_{i=1}^n (1+\sigma_i^2)^{-1/2}.$$

Therefore,

$$\left(\int_{\mathbb{R}^n} e^{-\frac{1}{2}\|x\|^2} \mu_n(dx)\right)^{-2} = \prod_{i=1}^n (1+\sigma_i^2) \geq 1 + \sum_{i=1}^n \sigma_i^2 = 1 + \mathrm{tr}(Q_n).$$

□

As we already completed the finite-dimensional calculation for the trace bound lemma, we now execute the limit to infinite dimensions and establish the canonical representation of Gaussian measures.

12. Completing the Trace Bound Proof: The Infinite Limit

Proof. From our finite-dimensional projection, we established the fundamental inequality:

$$1 + \operatorname{tr}(Q_n) = 1 + \sum_{i=1}^{n} \sigma_i^2 \leq \prod_{i=1}^{n} (1 + \sigma_i^2) = \left(\int_{\mathbb{R}^n} e^{-\frac{1}{2}\|x\|^2} \, \mu_n(dx) \right)^{-2}.$$

To generalize this to our infinite-dimensional Hilbert space H, take the limit as $n \to \infty$. First, observe the left side concerning the trace operator:

$$\lim_{n\to\infty} \operatorname{tr}(Q_n) = \lim_{n\to\infty} \sum_{i=1}^{n} \langle Qe_i, e_i \rangle = \sum_{i=1}^{\infty} \langle Qe_i, e_i \rangle = \operatorname{tr}(Q).$$

Simultaneously, evaluate the limit of the integral. By expanding the finite-dimensional norm using the inner products with our basis vectors, one gets

$$\int_H e^{-\frac{1}{2}\sum_{i=1}^{n} \langle x, e_i \rangle^2} \, \mu(dx) = \int_{\mathbb{R}^n} e^{-\frac{1}{2}\|x\|^2} \, \mu_n(dx).$$

Moreover,

$$\sum_{i=1}^{n} \langle x, e_i \rangle^2 \uparrow \|x\|_H^2,$$

so dominated convergence yields

$$\lim_{n\to\infty} \int_H e^{-\frac{1}{2}\sum_{i=1}^{n} \langle x, e_i \rangle^2} \, \mu(dx) = \int_H e^{-\frac{1}{2}\|x\|_H^2} \, \mu(dx) > 0.$$

Because this integral is strictly positive, its inverse square is finite, rigorously proving that the trace of Q is bounded and finite:

$$1 + \operatorname{tr}(Q) \leq \left(\int_H e^{-\frac{1}{2}\|x\|_H^2} \, \mu(dx) \right)^{-2} < \infty.$$

□

13. Spectral Properties of the Covariance Operator

Having established $\operatorname{tr}(Q) < \infty$, the finite trace property now allows the spectral decomposition of the covariance operator. Formalize the properties of Q via the following proposition.

Proposition 1. Let Q be a linear, continuous, symmetric, and positive semi-definite operator with $\operatorname{tr}(Q) < \infty$. By the spectral theorem for compact self-adjoint operators, there exists a Complete Orthonormal System (CONS), denoted as $(e_k)_{k\geq 1}$ in H, consisting entirely of the eigenvectors of Q. This means the action of Q is purely diagonalized on this basis:

$$Qe_k = \lambda_k e_k,$$

where the eigenvalues λ_k, which correspond to our variance components σ_k^2, are strictly non-negative:

$$\lambda_k \geq 0.$$

A critical topological consequence of this structure is that 0 is the only accumulation point of the eigenvalue sequence $(\lambda_k)_{k\geq 1}$. The finite trace property guarantees that the sum of these eigenvalues converges absolutely:

$$\operatorname{tr}(Q) = \sum_{k=1}^{\infty} \lambda_k < \infty.$$

14. The Abstract Fourier Series Foundation

To construct our Gaussian random variable, first recall the deterministic framework of the abstract Fourier series. Given the CONS $(e_k)_k$ in H, any arbitrary element $x \in H$ can be uniquely represented by the expansion

$$x = \sum_{k=1}^{\infty} \langle x, e_k \rangle e_k.$$

15. Canonical Representation of a Gaussian Measure

We now synthesize these components into a central proposition, the canonical representation, often referred to as the Karhunen–Loève expansion.

Proposition 2. Let $m \in H$, $Q \in L(H)$ as defined above, and let (e_k) be our specific CONS of eigenvectors. Let X be an H-valued random variable defined on a standard probability space $(\Omega, \mathcal{F}, \mathbb{P})$. The random variable X follows a Gaussian distribution,

$$X \sim N(m, Q),$$

if and only if it can be decomposed into the following infinite series:

$$X = \sum_{k=1}^{\infty} \sqrt{\lambda_k}\, \beta_k e_k + m.$$

In this representation, the coefficients β_k are independent and identically distributed standard normal random variables, meaning

$$\beta_k \sim N(0, 1).$$

For this representation to be valid, ensure the series converges. It is a strict property of this expansion that the series converges $\mathbb{P}$-almost surely. Furthermore, the convergence is strong enough to hold in L^p for all $p < \infty$.

As we already established the canonical representation for a static Gaussian random variable, we now dynamically extend this framework to define and construct the infinite-dimensional Q-Wiener process.

16. The One-Dimensional Projections of the Gaussian Variable

Given our Gaussian random variable $X \sim N(m, Q)$, project it onto our Complete Orthonormal System (e_k). The inner product $\langle X, e_k \rangle$ yields a one-dimensional Gaussian distribution:

$$\langle X, e_k \rangle \sim N(\langle m, e_k \rangle, \langle Q e_k, e_k \rangle).$$

Because e_k are the eigenvectors of Q, the variance term simplifies to the eigenvalue λ_k, allowing one to express this projection in distribution as

$$\sqrt{\lambda_k}\, N(0, 1) + \langle m, e_k \rangle.$$

This directly justifies our expansion of X in the Hilbert space:

$$X = \sum_{k=1}^{\infty} \langle X, e_k \rangle e_k,$$

which is equal in distribution to

$$\sum_{k=1}^{\infty} \sqrt{\lambda_k}\, N(0,1) e_k + \sum_{k=1}^{\infty} \langle m, e_k \rangle e_k.$$

17. Definition of the Q-Wiener Process

Let H and U be separable Hilbert spaces. Define a U-valued stochastic process $(W_t)_{t\in[0,T]}$ on a standard probability space $(\Omega, \mathcal{F}, \mathbb{P})$. This process is formally called a standard Q-Wiener process if and only if it satisfies three strict axioms:
(i) Initial Condition:

$$W_0 = 0.$$

(ii) Path Regularity: The mapping $t \mapsto W_t$ is $\mathbb{P}$-almost surely continuous, meaning it possesses continuous sample paths.
(iii) Independent Increments: For any arbitrary time partition

$$0 = t_0 < t_1 < \cdots < t_n \leq T,$$

the increments

$$W_{t_i} - W_{t_{i-1}}, \qquad i = 1, \ldots, n,$$

are mutually independent. Furthermore, each increment strictly follows a Gaussian distribution

$$N(0, (t_i - t_{i-1})Q).$$

18. Canonical Representation of the Wiener Process

We construct this Q-Wiener process explicitly via an infinite series expansion, analogous to our static variable. The canonical representation of W_t is

$$W_t = \sum_{k=1}^{\infty} \sqrt{\lambda_k}\, \beta_k(t) e_k.$$

In this representation, $\beta_k(t)$ are independent standard one-dimensional Brownian motions. For this representation to be mathematically rigorous, one must establish its convergence properties. The series converges $\mathbb{P}$-almost surely in the space of continuous functions $C([0,T], U)$, which guarantees uniform convergence with respect to time. Additionally, this convergence is strong enough to hold in

$$L^p(\Omega, \mathcal{F}, \mathbb{P}; C([0,T], U)) \qquad \text{for all } p < \infty.$$

19. Martingale Convergence and Doob's Inequality

To rigorously justify this uniform convergence over time, analyze the partial sums of this canonical series as a sequence of martingales, denoted $(M_t^n)_n$, where

$$M_t^n := \sum_{k=1}^{n} \sqrt{\lambda_k}\, \beta_k(t) e_k.$$

If this sequence of martingales forms a Cauchy sequence at the terminal endpoint T, Doob's inequality ensures that the sequence becomes uniformly close over the entire time interval $[0, T]$. Mathematically, if the sequence converges in L^p at time T, meaning

$$\lim_{n,m\to\infty} \mathbb{E}\big[\|M_T^n - M_T^m\|^p\big] = 0,$$

it strictly implies that the probability of the supremum of the deviation exceeding any ε goes to zero. One writes this final limit as

$$\lim_{n\to\infty} \mathbb{P}\left(\sup_{0\le t\le T} \|M_t^n - M_t\| > \varepsilon\right) = 0.$$

Appendix

A.1. Complete Orthonormal Systems

Definition 1 (Complete orthonormal system)**.** A family $(e_k)_{k\ge 1}$ in a Hilbert space H is called a complete orthonormal system if

$$\langle e_j, e_k\rangle = \begin{cases} 1, & j = k, \\ 0, & j \ne k, \end{cases}$$

and the linear span of the vectors (e_k) is dense in H.

A.2. Baire Category Theorem

Theorem 3 (Baire Category Theorem)**.** Let H be a complete metric space. If

$$H = \bigcup_{n\ge 1} F_n$$

with each F_n closed, then at least one of the sets F_n has non-empty interior.

A.3. Trace-Class Operators

Definition 2 (Trace-class operator)**.** Let $Q : H \to H$ be a bounded linear operator on a Hilbert space H. The operator Q is called trace class if

$$\operatorname{tr}(Q) = \sum_{k=1}^{\infty} \langle Qe_k, e_k\rangle < \infty$$

for one, and therefore for every, complete orthonormal system (e_k) of H.

A.4. Fatou's Lemma

Lemma 2 (Fatou's Lemma)**.** Let (f_n) be a sequence of nonnegative measurable functions on a measure space $(E, \mathcal{F}, \nu)$. Then

$$\int_E \liminf_{n\to\infty} f_n \, d\nu \le \liminf_{n\to\infty} \int_E f_n \, d\nu.$$

A.5. Riesz Representation Theorem

Theorem 4 (Riesz representation theorem)**.** Let H be a Hilbert space and let $L : H \to \mathbb{R}$ be a bounded linear functional. Then there exists a unique element $v \in H$ such that

$$L(g) = \langle v, g \rangle \qquad \text{for all } g \in H.$$

A.6. Dominated Convergence Theorem

Theorem 5 (Dominated convergence theorem)**.** Let (f_n) be a sequence of measurable functions on a measure space $(E, \mathcal{F}, \nu)$ such that $f_n \to f$ almost everywhere. If there exists an integrable function g with

$$|f_n| \leq g \qquad \text{for all } n,$$

then

$$\lim_{n\to\infty} \int_E f_n \, d\nu = \int_E f \, d\nu.$$

A.7. Standard Brownian Motion

Definition 3 (Standard Brownian motion)**.** A real-valued stochastic process $(\beta(t))_{t\in[0,T]}$ is called a standard Brownian motion if

$$\beta(0) = 0,$$

it has continuous sample paths, independent increments, and for all $0 \leq s < t \leq T$,

$$\beta(t) - \beta(s) \sim N(0, t - s).$$

A.8. Q-Wiener Process

Definition 4 (Q-Wiener process)**.** Let U be a separable Hilbert space and let $Q \in L(U)$ be symmetric and positive semidefinite. A U-valued process $(W_t)_{t\in[0,T]}$ is called a Q-Wiener process if

$$W_0 = 0,$$

its paths are almost surely continuous, it has independent increments, and

$$W_t - W_s \sim N(0, (t - s)Q) \qquad \text{for } 0 \leq s < t \leq T.$$

A.9. Doob's Maximal Inequality

Theorem 6 (Doob's maximal inequality)**.** Let $(M_t)_{t\in[0,T]}$ be a real-valued martingale and let $p > 1$. Then

$$\mathbb{E}\left[\sup_{0\leq t\leq T} |M_t|^p\right] \leq \left(\frac{p}{p-1}\right)^p \mathbb{E}[|M_T|^p].$$

Consequently, control of the terminal value M_T in L^p yields control of the supremum of the martingale over the whole interval.